

Retirement Obligations

To attract and retain talent, companies often offer retirement benefits to employees. These benefits include fixed pension payments, tax-advantaged savings plans, and promises to provide medical benefits when the employee retires. In some countries, companies are required to set up separate funds to pay these benefits, but inconsistencies are common because of differences in regulations and tax policy. For example, in the United States, companies must set up separate funds for pension promises (known as defined-benefit plans) but not for promises of retiree medical benefits. If the value of investments does not fully fund future promises, the company will have unfunded retirement obligations. Since the company is responsible for any shortfalls and these obligations take precedence over equity, any accurate valuation must account for them.

This chapter explores how to analyze and value a company with pension and other retirement obligations. Recent accounting changes have made the analysis easier, but careful thinking and reorganizing of financial statements are still required. The challenges include deciding which part of the pension expense is operating versus nonoperating, treating the balance sheet for unfunded or overfunded obligations, estimating the cost of capital for companies with pensions, and adjusting equity value to reflect unfunded (or overfunded) retirement obligations.

REORGANIZING THE FINANCIAL STATEMENTS WITH PENSIONS

In the past, accounting for pensions and other retirement obligations severely distorted operating profit, requiring adjustments to correctly measure the impact of retirement obligations on the company's value. In response, accounting policy has changed, gradually bringing the accounting for retirement obligations in line with the underlying doctrines of this text.

For companies that report under U.S. Generally Accepted Accounting Principles (GAAP), the changes occurred over many decades. Under original accounting principles, companies did not recognize unfunded pension liabilities on their balance sheets. The first changes involved recording unfunded retirement liabilities, albeit at a smoothed value intended to address the effects of short-term irregularities. In the 1980s, the rules were updated, and companies were required to record not only unfunded pension liabilities, but also other postretirement obligations, particularly promised medical benefits. Starting in 2006, companies were required to recognize the actual value of the unfunded (or overfunded) pension liability on the balance sheet.¹

Although the balance sheet reflected the value of unfunded pension liabilities after 2006, the pension expense continued to include both operating and nonoperating items. It included not only new benefits granted to employees, but also interest on the liability, returns on plan assets, and adjustments for actuarial changes. In 2018, nonoperating items were removed from pension expense. (This change mirrored adjustments we'd recommended in earlier editions of this book.) Any new benefits granted to employees are now included in the appropriate operating expense, such as cost of sales or selling expenses. All other items, such as interest expense, actuarial changes, and earnings, are classified as "other" expense.

Because of these changes, you no longer need to make as many adjustments to the GAAP financial statements for pensions and retirement benefits for 2018 and forward. You must still adjust financial statements released before 2018. International Financial Reporting Standards (IFRS) have been updated as well, but under multiple revisions to existing standards, rather than by adopting an entirely new standard.

Throughout the chapter, we examine pension accounting using the American food manufacturer Kellogg. Kellogg is an interesting case because, even though its pension obligations are almost fully funded (about 95%, which is within the margin of year-to-year fluctuations), it's still necessary to dive into the details to make the right adjustments to estimate net operating profit after taxes (NOPAT) and invested capital. The information required to analyze pensions for Kellogg is in the footnotes to their financial statements, note 10, "Pension Benefits," and note 11, "Nonpension Postretirement and Postemployment Benefits."² In these two notes, the company provides information on projected benefit obligations, the fair value of plan assets, and a breakout of the annual pension expense.

Reorganizing the Balance Sheet

To start, find all the retirement-related assets and liabilities on the balance sheet. If these items are relatively small, companies may include prepaid pension

¹ Statement of Financial Accounting Standards (SFAS) 158 was passed by the Financial Accounting Standards Board (FASB) in September 2006. Accounting Standards Update (ASU) Number 2017-07 was passed by the FASB in March 2017.

² All the data related to Kellogg in this chapter appear in Kellogg's 2018 10-K filing.

EXHIBIT 23.1 Kellogg: Pension Note in Annual Report, Funded Status

\$ million

	Pension benefits ¹	Other benefits ²	Total benefits
Fair value of plan assets at end of year	4,677	1,140	5,817
Projected benefit obligation at end of year	(5,117)	(1,069)	(6,186)
Funded status	(440)	71	(369)
Amounts included in the consolidated balance sheet			
Other assets	228	107	335
Other current liabilities	(17)	(2)	(19)
Pension liability	(651)	—	(651)
Other liabilities	—	(34)	(34)
Net amount recognized	(440)	71	(369)

¹ Kellogg 2018 annual report, Note 10, "Pension Benefits."² Kellogg 2018 annual report, Note 11, "Nonpension Postretirement and Postemployment Benefits."

assets in other long-term assets and unfunded pension liabilities as part of other long-term liabilities, but the details will be in the pension footnote.

Exhibit 23.1 reports the funded status of Kellogg's defined-benefit plans and the location of the company's underfunding on the balance sheet, as reported in the notes. In 2018, Kellogg had \$369 million in unfunded pension and other postretirement liabilities. This amount does not appear as a single value on the balance sheet. Instead, the net underfunding is disaggregated across four accounts, including \$335 million embedded in other assets, \$19 million embedded in other current liabilities, a pension liability of \$651 million, and \$34 million embedded in other liabilities. A company can have both excess pension assets and unfunded pension liabilities, because companies may have multiple pension plans, and pension assets from one plan are not netted against underfunding from another.

Note that most companies don't fund their "other" retirement obligations, like promised medical benefits, so this will typically appear as showing zero assets and only the liability.

When reorganizing the balance sheet, separate operating assets from pension assets, and treat excess pension assets as nonoperating. Unfunded pension liabilities (on a gross basis) should be treated as a debt equivalent and, as such, should not be deducted from operating assets to determine invested capital. Instead, they will be valued separately during the transition from enterprise value to equity value.

Reorganizing the Income Statement

Pension accounting combines several items into a single expense, known as the pension expense. Some components are operating, while others are related to the performance of the plan assets. As such, pension expense must be

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EXHIBIT 23.2 **Kellogg: Pension and Other Postretirement Expenses**

\$ million				
	2016	2017	2018	
Service cost	119	114	105	→ Operating expense
Amortization of prior service cost	4	—	(1)	→ Operating for historical benchmarking only
Interest cost	213	201	201	→ Nonoperating, related to plan performance
Expected return on plan assets	(442)	(469)	(455)	
Recognized net (gain) loss	304	(126)	350	
Settlements (curtailments)	1	(151)	(30)	→ Information recorded on income statement
Net periodic (benefit) cost	199	(431)	170	

Source: Kellogg 2016–2018 annual reports.

analyzed line by line. Exhibit 23.2 presents the pension expense for Kellogg. For ease of exposition, the exhibit combines pension expense with other postretirement benefits, which Kellogg reports in two separate notes.

In Exhibit 23.2, you will find six accounts. Service cost and the amortization of prior service cost represent benefits granted to the employee in return for service to the company.³ Interest cost on plan liabilities, expected return on plan assets, and recognized gains and losses represent the evolution of plan assets and liabilities over time.⁴ If the change in plan assets matched the change in plan liabilities each year, these accounts would cancel. Since markets are volatile, this is not the case. As a result, the investment performance of plan assets contaminates pension expense. Curtailments represent changes to the pension plan that restrict benefits.

Prior to 2018, companies using GAAP reported the entire pension expense as part of operating expenses. Although not visible on the income statement, the expense was subtly embedded in cost of sales and in selling, general, and administrative (SG&A) expenses. This meant operating expenses and consequently operating profit were a function of the investment performance of plan assets, leading to distortions in competitive benchmarking.

To better understand potential distortions, examine the portion of pension expense titled “recognized net (gain) loss” in Exhibit 23.2. In 2016, Kellogg recognized \$304 million in *losses* on plan assets. This increased pension expense relative to other years. In 2017, Kellogg reported recognized *gains* of \$126 million. This caused pension expense to convert from a \$199 million expense in 2016 to a \$431 million benefit in 2017. Since pension expense is embedded within cost of sales, this caused

³ Service cost represents the present value of retirement promises given to the company’s employees in a particular year. Prior service costs are additional retroactive benefits given to employees from an amendment to the pension plan. Prior service costs are not expensed immediately. Instead, they are amortized over the expected lifetimes of employees. For more on pension accounting, see D. Kieso, J. Weygandt, and T. Warfield, *Intermediate Accounting*, 17th ed. (Hoboken, NJ: John Wiley & Sons, 2019), chap. 20.

⁴ Interest cost represents the present value of service cost growing into the actual retiree payout. Expected return on plan assets equals the expected return based on asset mix. Recognized gains and losses represent the gradual recognition of past gains and losses of the pension fund.

EXHIBIT 23.3 Kellogg: EBITA Adjusted for Pensions

\$ million

	2016	2017	2018
Operating profits, unadjusted			
Revenues	13,014	12,923	13,547
Operating costs	(11,619)	(10,977)	(11,841)
EBITA, unadjusted	1,395	1,946	1,706
Operating profits, adjusted			
Revenues	13,014	12,923	13,547
Operating costs	(11,619)	(10,977)	(11,841)
Add: Net periodic (benefit) cost	199	(431)	—
Less: Service cost	(119)	(114)	—
EBITA, adjusted	1,475	1,401	1,706
Operating margin, %			
Operating margin, unadjusted	10.7	15.1	12.6
Operating margin, adjusted	11.3	10.8	12.6

Source: Kellogg 2016–2018 annual reports.

operating profit to rise in 2017. As a result, the unadjusted operating margin rose from 10.7 percent to 15.1 percent in 2017, even though adjusted margins—that is, those that only include service expense—fell from 11.3 percent to 10.8 percent.

To eliminate plan performance from past operating expenses, remove the pension expense—in Kellogg’s case, a \$431 million gain in 2017—and replace it with the service cost of \$114 million. These adjustments are shown in the middle section of Exhibit 23.3. If benchmarking across companies, one can also include the amortization of prior service costs as an operating expense. While these prior service costs represent real benefits given to employees, they are noncash and represent past changes. Therefore, they should not be incorporated into free cash flow. These service costs are also often small and unlikely to change our perception of historical performance; for simplicity’s sake we usually treat them as nonoperating.

Under new accounting standards, the pension expense is no longer treated as operating. Instead, service cost is allocated to the appropriate operating expense account (cost of goods sold or SG&A), and the nonoperating portion of pension expense is treated as “other income or expenses,” as is the case for Kellogg in 2018. Since other income or expenses usually accumulates nonoperating items, this makes reorganizing simple.

Given that accounting guidelines across countries may differ, always check the notes for the location of various elements and adjust accordingly.

Expected Return and Earnings Manipulation

To avoid volatility in the income statement, accounting standards allow companies to include an “expected return” on pension plan assets as part of pension expense, rather than actual returns. For example, Exhibit 23.2 shows that Kellogg recorded \$455 million in expected return on plan assets in 2018, even

EXHIBIT 23.4 **Capital Structures of Three Consumer Products Companies, 2018**

\$ million

	Kellogg	General Mills	Mondelēz
Projected benefit obligations ¹	6,186	7,415	11,089
Value of plan assets ¹	(5,817)	(6,904)	(9,975)
Unfunded pension liabilities	369	511	1,114
Debt, net of cash	9,221	16,225	19,826
Debt and debt equivalents	9,590	16,736	20,940
Market value of equity	19,784	25,073	58,197
Enterprise value	29,374	41,809	79,137

¹ Includes domestic pensions, foreign pensions, and other retirement obligations.

Source: Kellogg, General Mills, and Mondelēz 2018 annual reports.

though the plan assets lost \$350 million that same year. This enables companies to smooth pension returns from year to year, avoiding volatility in net income.

Since expected return must be estimated, company management has discretion over the rate used—a license that management may sometimes use to manipulate accounting profitability. Bergstresser, Desai, and Rauh found that management increases expected rates of return to increase profitability immediately before acquiring other firms and before exercising stock options.⁵ They also found that companies with the weakest shareholder protections tend to use the highest estimates for expected return. With nonoperating items now incorporated into other income and expenses, cost of sales and operating profit are no longer affected by expected-return choices. Even so, net income remains susceptible, which is just one of many reasons why NOPAT, and not earnings per share (EPS), remains a critical measure of accurate benchmarking and financial forecasting.

PENSIONS AND THE COST OF CAPITAL

A key component of valuation is the cost of equity, which is typically estimated using the capital asset pricing model (CAPM) and beta. As discussed in Chapter 15, it is difficult to accurately measure the beta of a single company. Therefore, we recommend using an industry beta derived from multiple companies in similar lines of business.

To isolate the economic risk each company faces, it is important to remove the effect of leverage and, if meaningful, the effect of pensions from beta. In Exhibit 23.4, we present the pension details, debt, and equity of three consumer

⁵ D. B. Bergstresser, M. A. Desai, and J. Rauh, "Earnings Manipulation, Pension Assumptions, and Managerial Investment Decisions," *Quarterly Journal of Economics* 121, no. 1 (February 2006): 157–195. For more on shareholder protection indexes, see P. Gompers, J. Ishii, and A. Metrick, "Corporate Governance and Equity Prices," *Quarterly Journal of Economics* 118, no. 1 (2003): 107–155.

products companies, including Kellogg. The data include pension plans and other retiree benefits, such as health care. Each company's plan is well funded, with pension shortfalls at or below 10 percent of projected benefit obligations.

There are two ways to incorporate pensions into the unlevering process. In the first method, we assume the pension fund manager has successfully matched the beta risk of plan assets to the beta risk of projected benefits. In this case, the funded portion will net out, and only the unfunded portion will affect the equity beta. In the second method, we relax the assumption of matched beta. While the second method is more flexible than the first, it requires an estimate of the beta risk for plan assets. Since the estimate requires data found only in the notes (versus a professional data provider), as well as a few assumptions regarding asset composition, its use should be limited to situations where pensions play a critical role in company valuation.

In the first method, we assume that only the unfunded pension liability affects the equity beta. Since the unfunded pension liability mirrors debt, we can use the equation for unlevering beta presented in Chapter 15:

$$b_u = \frac{D}{V} b_d + \frac{E}{V} b_e \quad (1)$$

where b_u equals the unlevered beta, b_d equals the beta of debt, b_e equals the beta of equity, and E equals the market value of equity. The unfunded pension liability is a debt equivalent. Therefore, D equals traditional debt *plus* unfunded pension liabilities less excess cash.

In Exhibit 23.5, we estimate the unlevered beta for Kellogg and two other companies. We present the results with and without pensions for the purpose of comparison. In the analysis, we assume a debt beta of 0.17. Many assume that the debt beta equals zero, but we use a positive beta to assess the various methodologies in a consistent manner. The beta of equity for Kellogg, measured using five years of monthly stock returns, equals 0.64. The debt-to-value

EXHIBIT 23.5 **Unlevered Betas for Three Consumer Products Companies**

	Kellogg	General Mills	Mondelēz	
Beta of debt	0.17	0.17	0.17	
Beta of equity ¹	0.64	0.75	0.83	
Beta of plan assets ²	0.66	0.75	0.42	
Debt-to-value, excluding pensions, %	31.8	39.3	25.4	
Debt-to-value, including pensions, %	32.6	40.0	26.5	
Unlevered beta				Average
Unlevered beta, unadjusted for pensions	0.49	0.52	0.66	0.59
Method 1: Treat unfunded pension as debt equivalent	0.48	0.52	0.66	0.59
Method 2: Allow plan asset beta to differ from obligations beta	0.39	0.42	0.63	0.52

¹ Beta of equity from ThomsonOne, July 2019.

² Assumes the beta of debt investments equals 0.17 and the beta of all remaining investments equals 1.0.

ratio equals 31.8 percent without unfunded pensions and 32.6 percent with unfunded pensions. The resulting unlevered betas with and without unfunded pensions are nearly identical because Kellogg's unfunded pension of \$369 million is quite small compared with its debt of \$9.2 billion. Not surprisingly, the equity beta is higher than the unlevered beta, because leverage increases risk.

To unlever beta when the risk is mismatched, we separate plan assets from pension liabilities and apply the teachings of economists Franco Modigliani and Merton Miller (see Chapter 10) to solve for the risk of operating assets. In Appendix C, we step through the algebraic derivation, leading to the following formula for unlevered beta:

$$b_u = \frac{D + V_{pbo}}{V} b_d + \frac{E}{V} b_e - \frac{V_{pa}}{V} b_{pa} \quad (2)$$

where b_u represents unlevered beta, b_d represents the beta of debt, b_e represents the equity beta, b_{pa} represents the beta of plan assets, D equals the value of traditional debt net cash, V_{pbo} equals the projected benefit obligations, E equals the market value of equity, V_{pa} equals the market value of plan assets, and V equals enterprise value, as measured by the sum of debt, unfunded pension liabilities, and the market value of equity.

To measure the beta of plan assets, we use the target allocation reported in the pension footnote. Following the research of Jin, Merton, and Bodie, we assume debt securities have a beta of 0.17 and other investments have a beta of 1.0.⁶ Using the data from Exhibit 23.4 and Exhibit 23.5, we solve for unlevered beta. For Kellogg, the resulting unlevered beta equals 0.39. Because the beta of plan assets exceeds that of projected benefits, the resulting estimate of unlevered beta is lower than we obtained using earlier methods. Had the betas for plan assets and plan liabilities been the same, the two methods would yield the same results.

While each method has its benefits, we believe Equation 1 inclusive of unfunded pension liabilities is the easiest and most reliable method for unlevering beta. An estimate of unfunded pension liabilities is already required for equity valuation, and the method does not require an extensive analysis of plan assets—a daunting task when there are many companies to analyze within an industry.

RELEVERING BETA TO ESTIMATE THE COST OF EQUITY

Once you have estimated the unlevered industry beta, relever the industry beta to the company's target capital structure and compute the company's cost of capital. To relever the industry beta, do *not* incorporate pensions. While

⁶ L. Jin, R. Merton, and Z. Bodie, "Do a Firm's Equity Returns Reflect the Risk of Its Pension Plan?" *Journal of Financial Economics* 81, no. 1 (2006): 1–26.

this may seem inconsistent for a company with pensions, it is not. We have eliminated pensions from free cash flow and the cost of capital, and there is no reason to reintroduce pensions, or the risk associated with them, into the value of operations. Instead, value pensions separately, and sum the parts.

INCORPORATING PENSIONS INTO THE VALUE OF EQUITY

Pension plans and other obligations, such as promised medical benefits, will affect a company's value in two ways. First, service cost will be embedded within free cash flow. Since only cash contributions and not service costs are tax deductible, make sure to adjust taxes appropriately for companies that systematically underfund their obligations. Not every country provides tax relief on pension contributions, so check local tax law to determine the marginal tax rate for contributions. Second, past over- or underfunding must be incorporated into value as a nonoperating asset or debt equivalent.

For an ongoing enterprise, excess pension assets can be netted against unfunded liabilities to determine net assets (or liabilities) outstanding. If the company is being valued for liquidation or the pension plan is being terminated, net unfunded liabilities cannot be netted against excess pension assets, as most countries charge a significant penalty for withdrawing excess funds from pension plans. Instead, add after-tax excess pension assets at the penalty rate, and deduct after-tax unfunded pension liabilities at the marginal tax savings for pension contributions.

To value companies with net unfunded liabilities, reduce enterprise value by the product of $(1 - \text{marginal tax rate})$ times net pension liabilities. To incorporate pensions for a company with net excess assets, increase enterprise value by the product of $(1 - \text{marginal tax rate on pensions})$ times net pension assets, as excess pension assets will lead to fewer required contributions in the future.

In 2018, Kellogg recognized \$440 million in unfunded pension liabilities and \$71 million in prefunded other benefits (see Exhibit 23.1), for a net total liability of \$369 million. Assuming a marginal tax rate of 24 percent, the after-tax liability equals \$280 million. To determine equity value, deduct the after-tax liability from enterprise value.

CLOSING THOUGHTS

The International Accounting Standards Board and the U.S.-based Financial Accounting Standards Board have worked to eliminate the distortions caused by pension accounting. For most companies, the income statement now separates service cost from nonoperating pension expenses, and the balance sheet recognizes the market value of unfunded pension obligations. The result is better benchmarking, requiring fewer adjustments, and a valuation that is easier to carry out.

REVIEW QUESTIONS

1. How did GAAP standards for pension accounting change in 2018? What impact does this change have on an analyst's estimation of key performance metrics for U.S. companies?
2. If there is no line item for prepaid pension assets or unfunded pension liabilities on a firm's balance sheet, does this mean the firm's pension plan is fully funded? Explain.
3. AutoCo, a large automobile company, reports operating income of \$15.0 billion, \$15.5 billion, and \$15.7 billion over the next three years on sales of \$40.0 billion, \$41.0 billion, and \$43.0 billion. Exhibit 23.6 displays a breakout of its pension expense from the footnotes of its annual report. Use this breakout to eliminate nonoperating income related to pensions from operating income reported in AutoCo's income statement. What is the impact of adjusting operating income on operating margins?

EXHIBIT 23.6 **AutoCo Annual Report****AutoCo: Annual report, Note 10**

Postretirement benefits and employee stock ownership plan

	Pension benefits			Other retiree benefits		
	2016	2017	2018	2016	2017	2018
Service cost	210	270	280	90	95	85
Interest cost	450	520	500	250	230	220
Expected return on plan assets	(500)	(600)	(650)	(450)	(500)	(550)
Prior service cost (credit) amortization	10	20	10	(30)	(20)	(30)
Recognized net (gain) loss	(150)	(200)	(250)	5	10	5
Curtailment and settlement gain	10	(40)	(200)	—	(5)	(5)
Gross benefit cost (credit)	30	(30)	(310)	(135)	(190)	(275)

4. How might managers manipulate earnings through the use of retirement expenses?
5. What are the two methods to incorporate pensions into the cost of capital estimation? Under what circumstances is one method preferred over the other?
6. In relevering a firm's equity beta, why should you exclude pensions from the calculation?
7. How should you incorporate pensions into a firm's equity value?